

Week 1 - Day 3b Supplemental: For Loops

This section contains extra material related to for loops.

Exercise 1: Dictionary Zipping[]

Given an empty dictionary `dic`, and two equal-length lists `keys` and `vals`. Put every corresponding key and value from `keys` and `vals` (with the same index) into the dictionary `dic`. For example, if the list are `keys = ["Jelani", "Daniel"]` and `vals = ["Week 4", "Week 1"]`, then your dictionary `dic` should contain two keys, and `dic["Jelani"] = "Week 4"`, `dic["Daniel"] = "Week 1"`.

```
# TODO: Your code here
keys = ["Jelani", "Daniel"]
vals = ["Week 4", "Week 1"]
```

Exercise 2: Factors []

2.1 For a given number `n`, use a `for` loop to print a list contains all the factors of `n`. For example, for `n = 20`, your output should be `[1, 2, 4, 5, 10, 20]`.

```
# TODO: Your code here
```

Exercise 3: Patterns []

3.1 Use a `for` loop to print the following: `$\nnewline * \nnewline * \\ * \nnewline * \\ * \\ * \nnewline * \\ * \\ * \\ * $`

```
# TODO: Your code here
```

3.2 Use a nested `for` loop to print the following: `$\nnewline 1 \nnewline 2 \\ \\ 3 \nnewline 4 \\ \\ 5 \\ \\ 6 \nnewline 7 \\ \\ 8 \\ \\ 9 \\ \\ 10$`

```
# TODO: Your code here
```

3.3 Use a nested `for` loop to print the following: `$\nnewline 7 \\ \\ 8 \\ \\ 9 \\ \\ 10 \nnewline 4 \\ \\ 5 \\ \\ 6 \nnewline 2 \\ \\ 3 \nnewline 1$`

```
# TODO: Your code here
```

Exercise 4: Matrices []

Print out the following. `$ \nnewline 11 12 13 14 15 \nnewline 20 19 18 17 16 \nnewline 21 22 23 24 25 \nnewline 30 29 28 27 26 \nnewline 31 32 33 34 35$`

4.1 Use a nested `for` loop to solve it

```
# TODO: Your code here
```

4.2 Use just one `for` loop to solve it

```
# TODO: Your code here
```